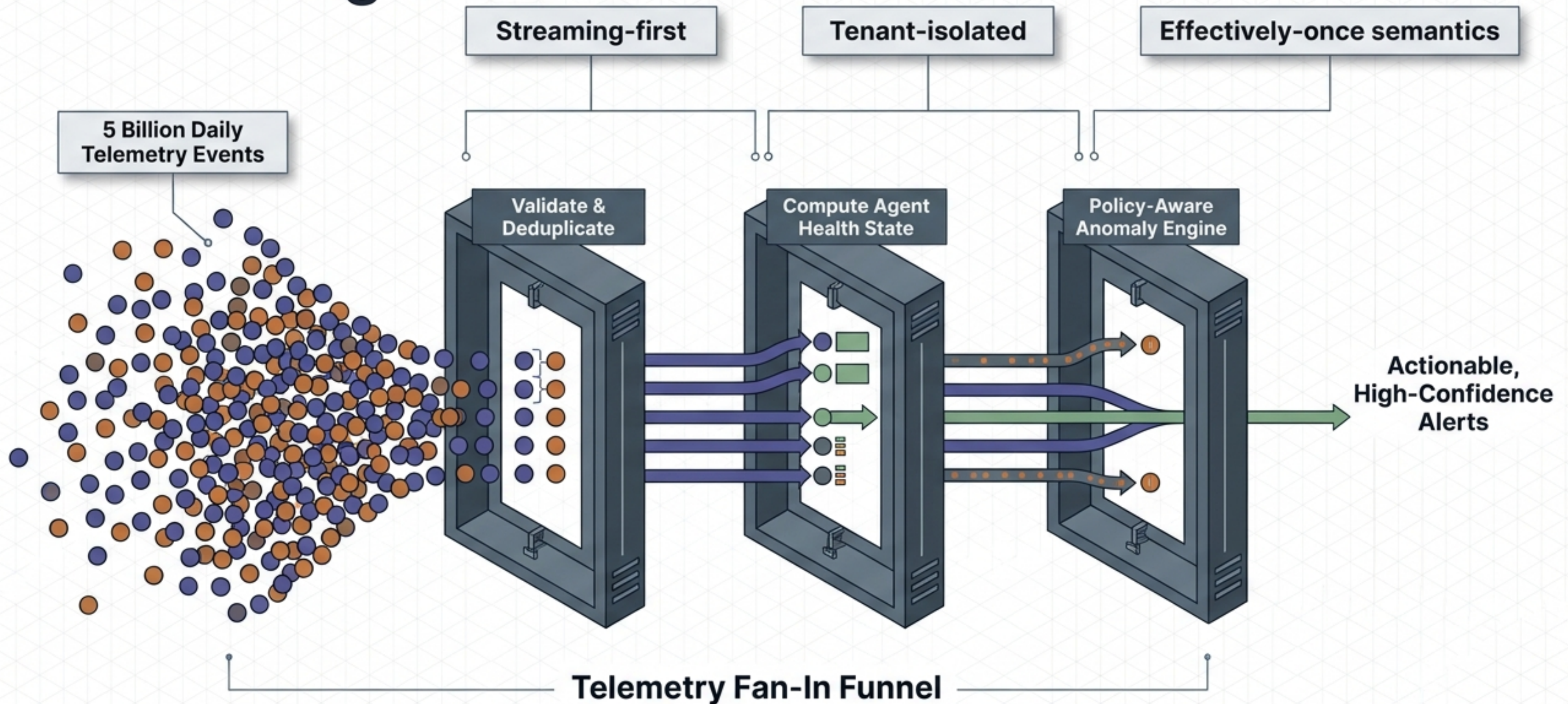


Architecting Trust at SentinelOne Scale

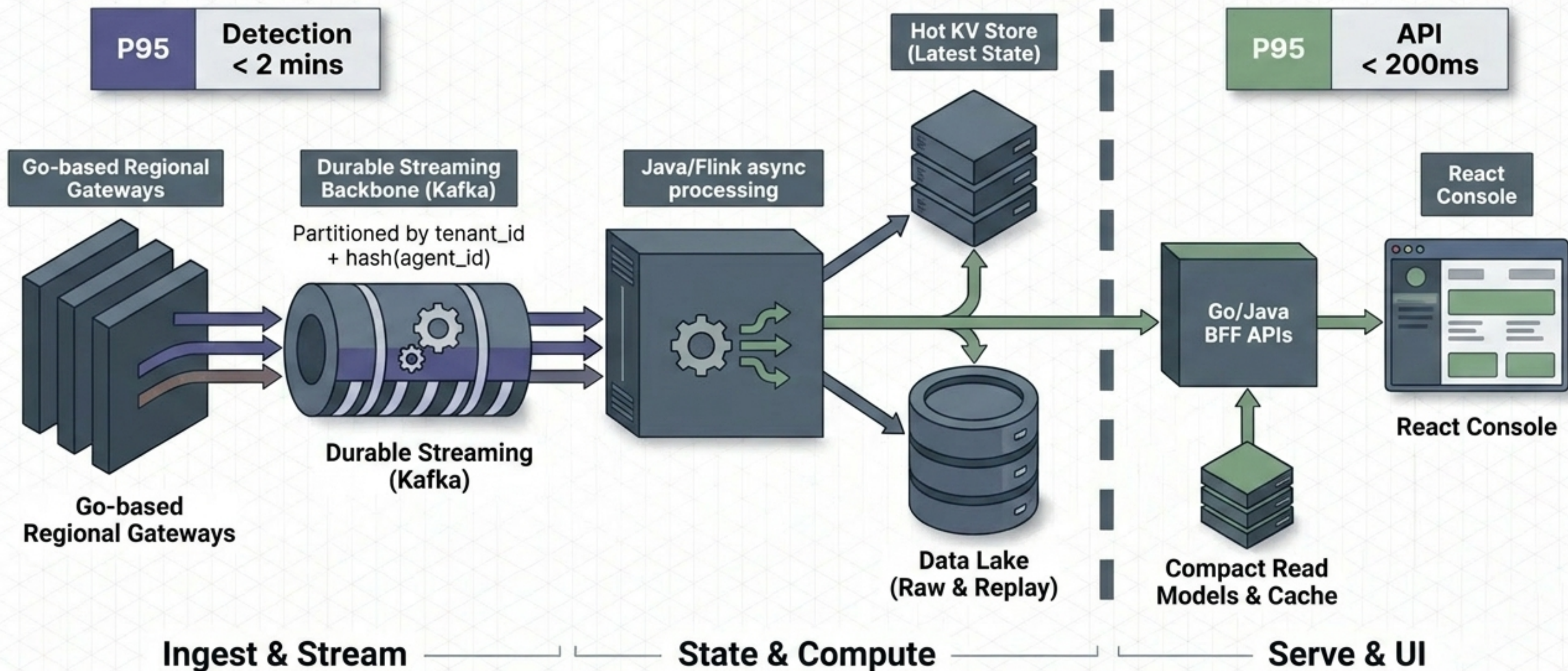
An Engineering and Execution Blueprint for the **Singularity Health Centre**.



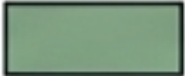
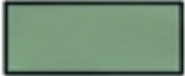
Protecting customer trust over alerting volume



Isolating high-volume writes from low-latency reads



Disagree and Commit: Resolving the alert coalescing tradeoff

Technical Tradeoff Matrix	Engineer A: Stream-Time Coalescing (Kafka/Flink)	Engineer B: DB Writes + Async Cron Jobs
Latency SLO (< 5 min)	Strong 	Risky 
Write Amplification	Optimized	Expensive hot tables 
Operational Simplicity	Steep learning curve	Highly familiar
Failure Isolation	Strong 	Silent cron failures possible

Streaming Coalescing Selected. We protect the detection SLO and prevent notification spam before customer impact, preserving raw events in object storage.

Missing data does not equal endpoint failure

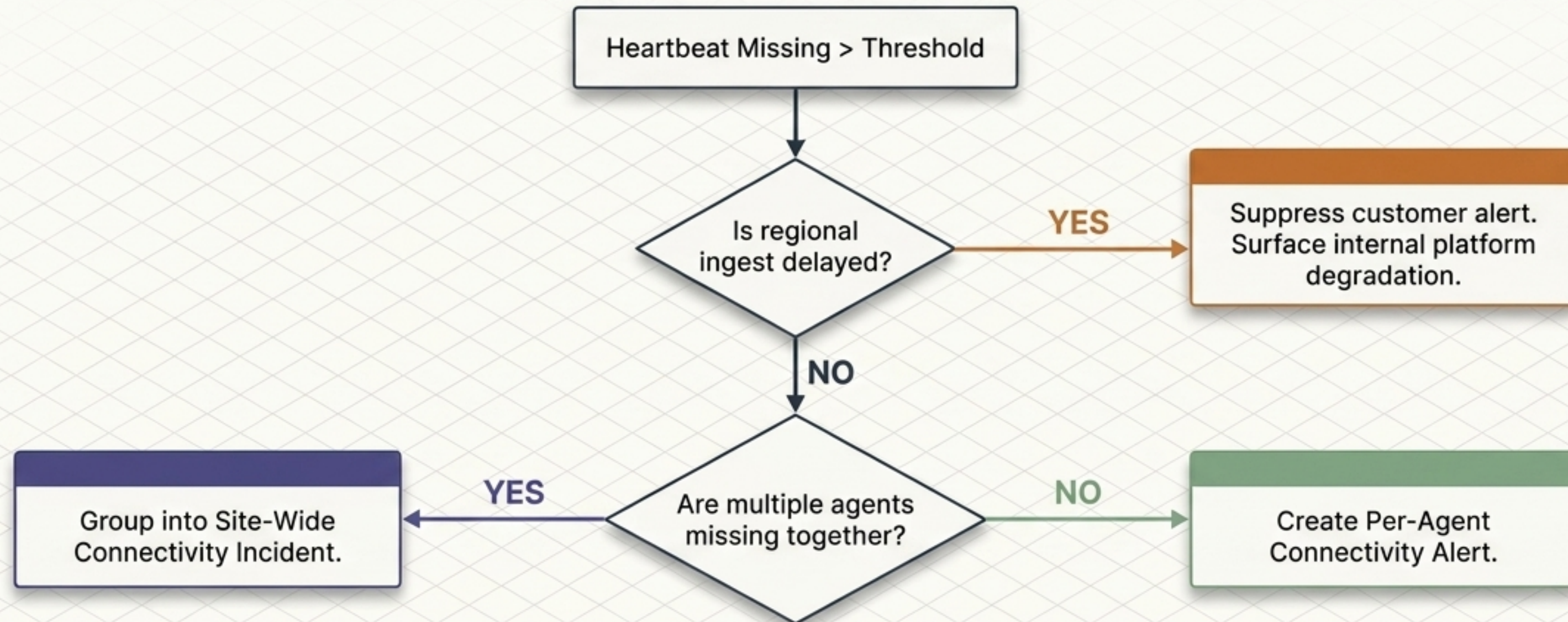


Endpoint
Heartbeat Age



Regional Platform
Freshness

The Freshness Correlation Engine



Defining the boundary between proven value and production safety

MVP – Prove Value

Consume central gateway telemetry

Compute health state

Core anomalies (disabled, anti-tamper, connectivity, low disk)

Shadow mode validation

GA – Replace Legacy Safely

Multi-region scale

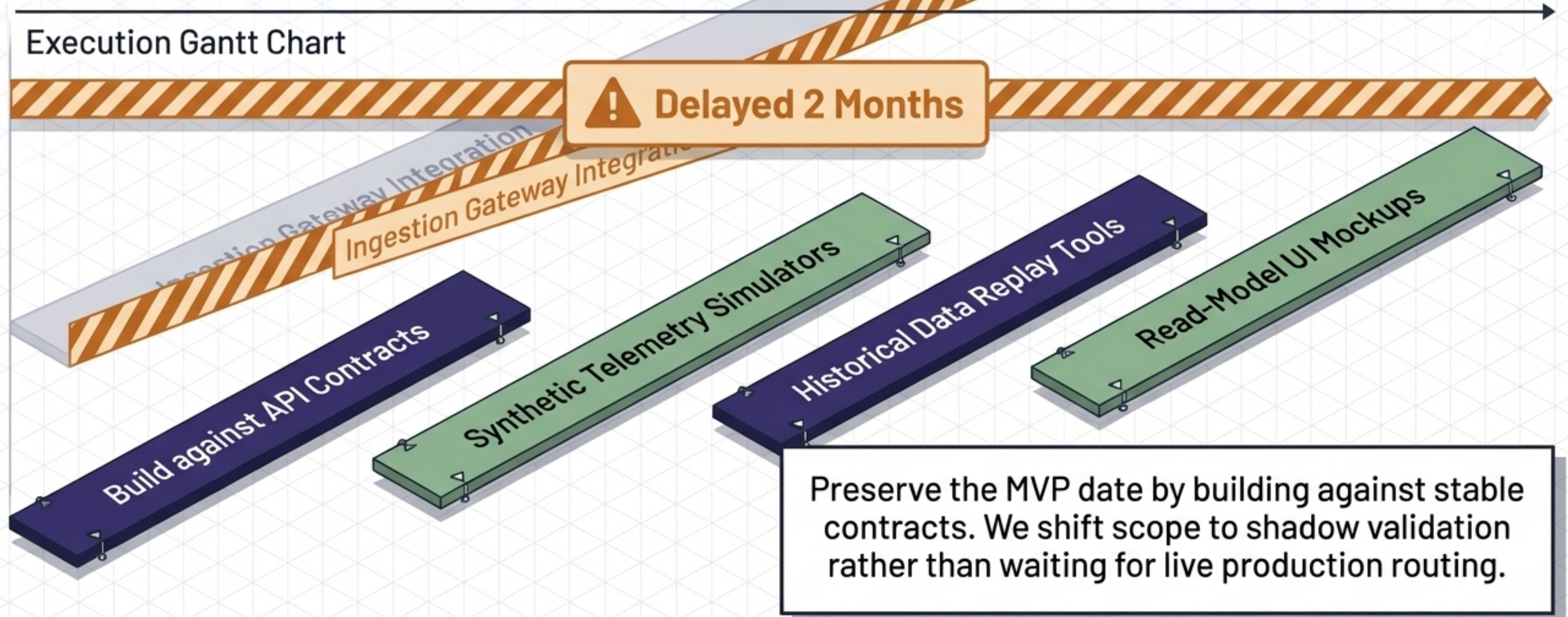
Alert grouping

Full RBAC & maintenance windows

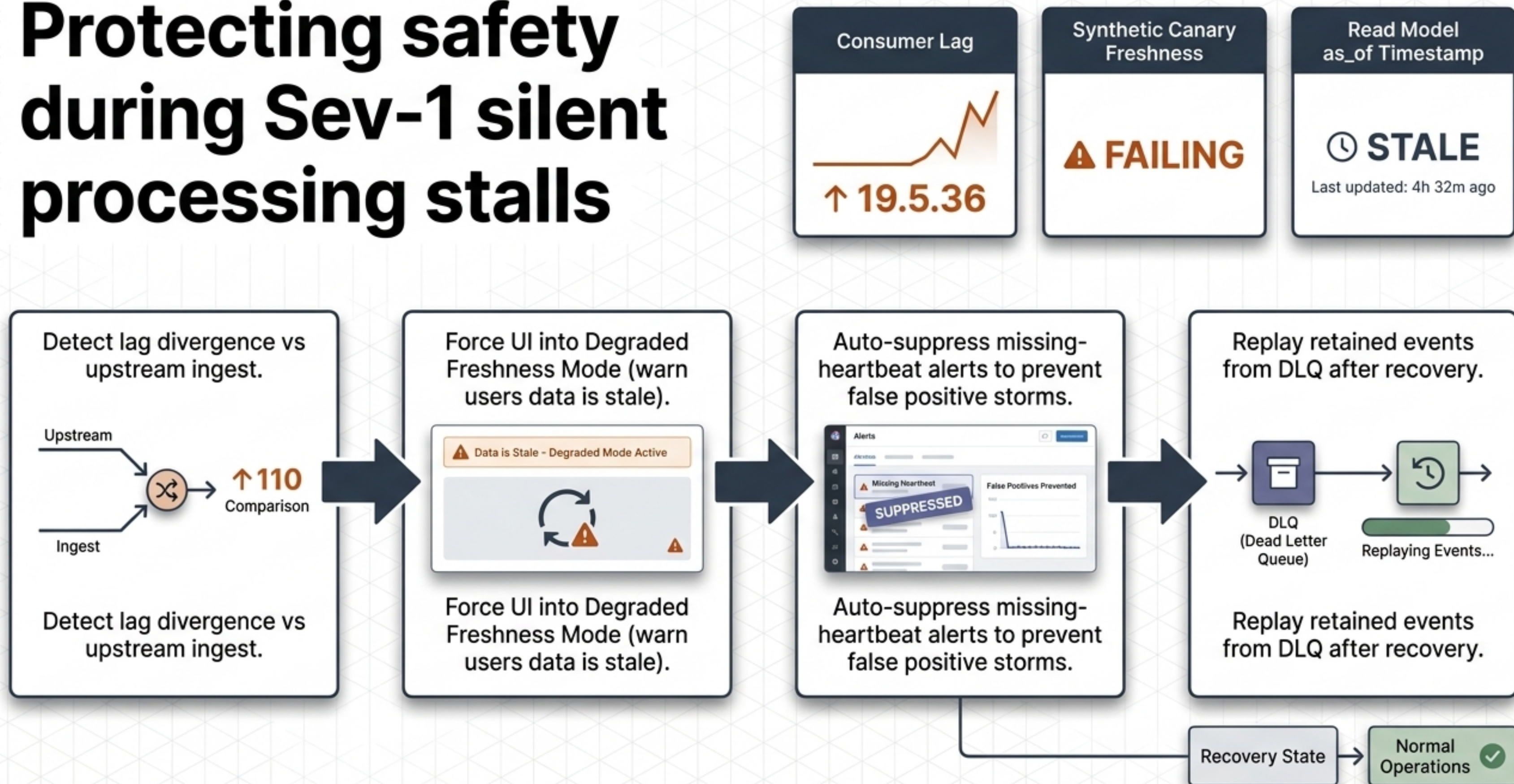
Legacy offline-status deprecation

MVP rules must pass cross-functional false-positive reviews before exiting shadow mode.

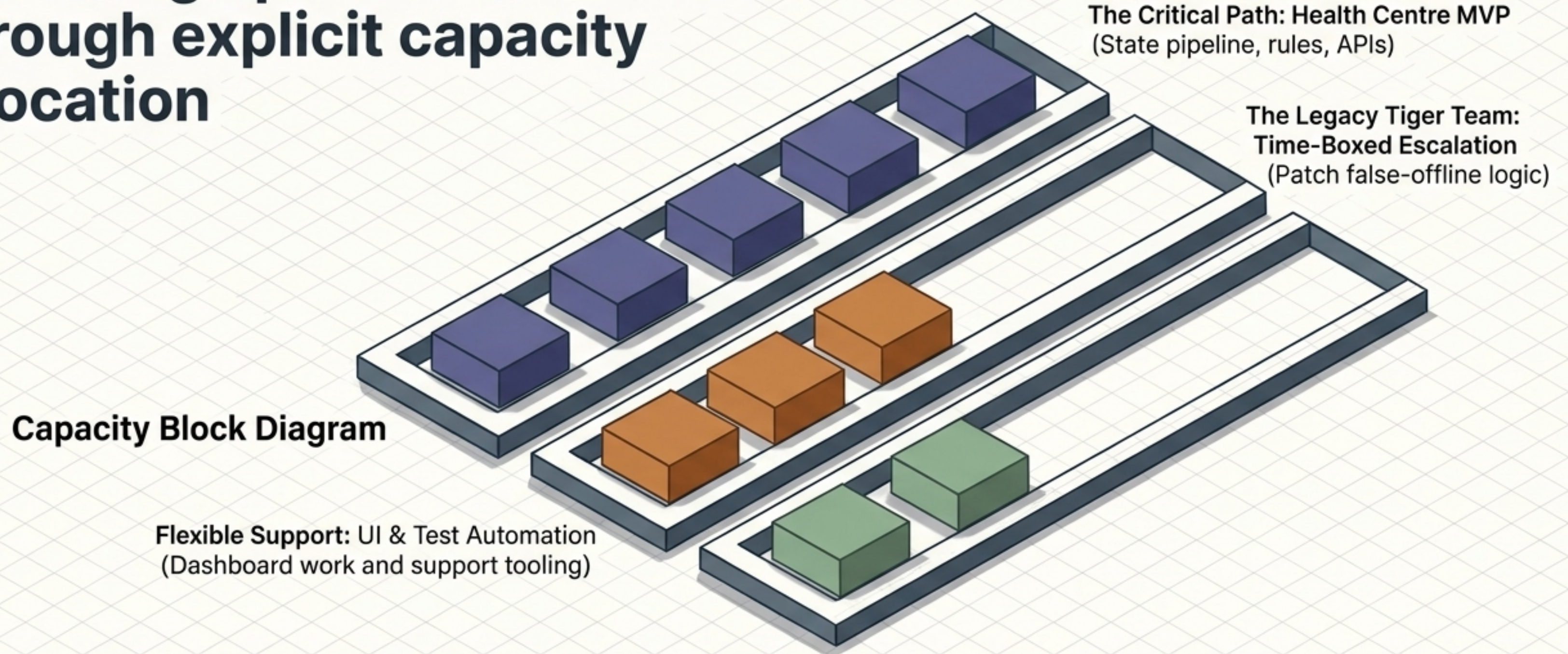
Advancing the critical path despite blocked dependencies



Protecting safety during Sev-1 silent processing stalls

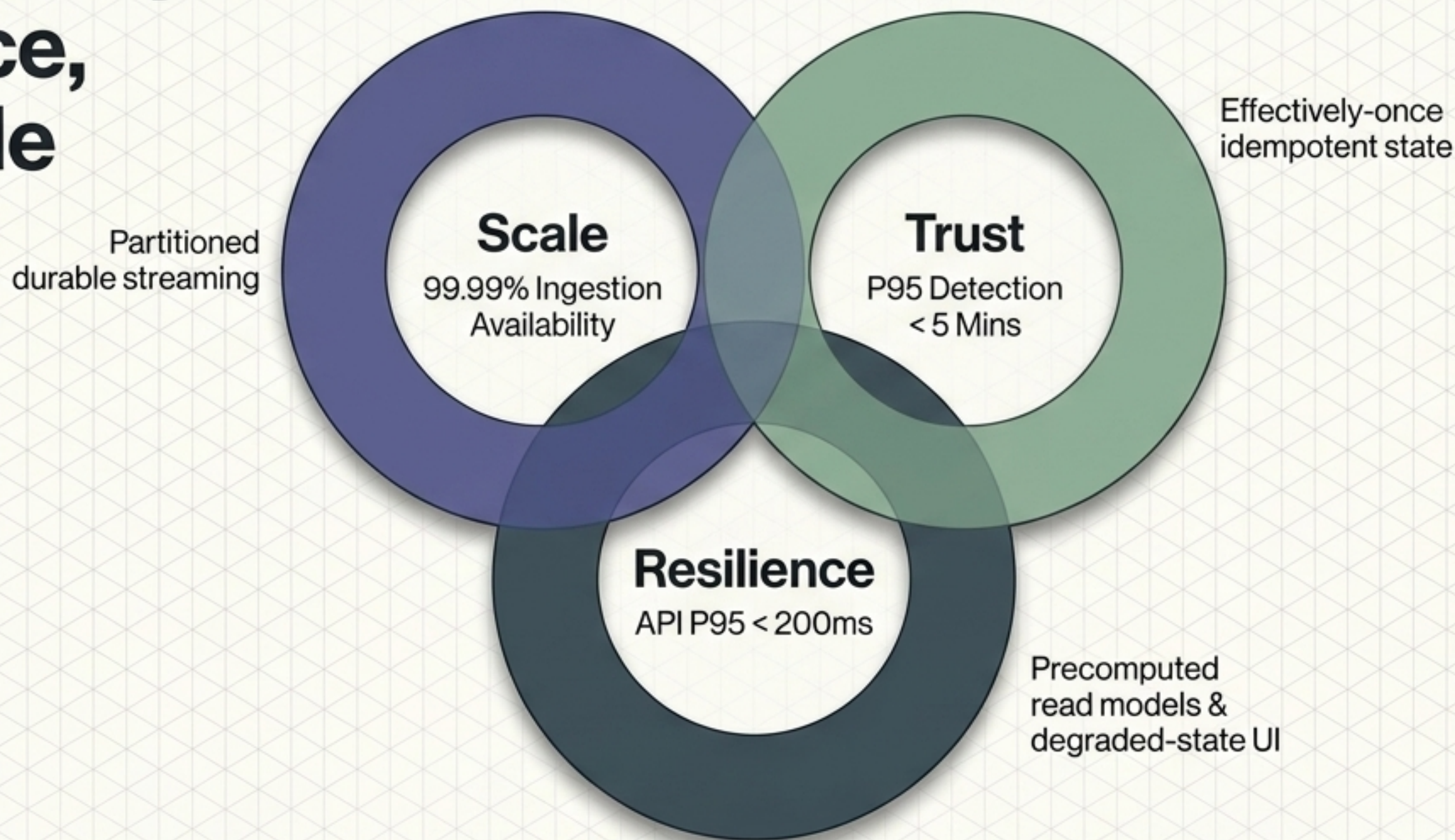


Bounding operational drains through explicit capacity allocation



We do not sacrifice current customer trust to ship the future. We patch the acute issue, then use the learnings to accelerate Health Centre migration.

A health platform measured by trust, resilience, and scale



Singularity Health Centre: Streaming-first. Replayable. Radically transparent.